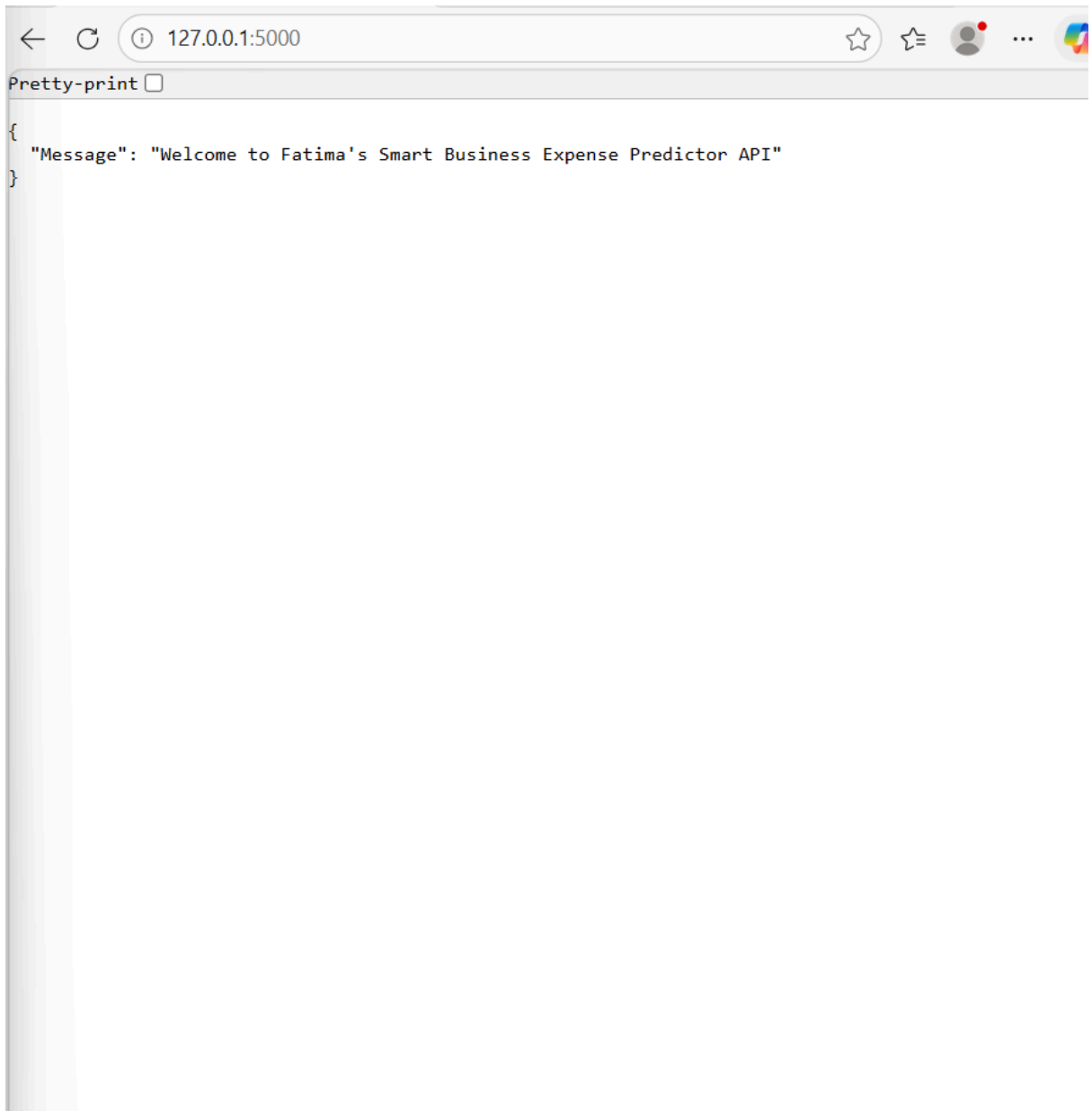


```
Edit Selection Find View Goto Tools Project Preferences Help
app.py — flask_project  generate_data.py  app.py — expense_predictor_api  train_model.py  home.html
1 from flask import Flask, jsonify, request, render_template
2 import joblib
3 import numpy as np
4 import os
5
6 app = Flask(__name__)
7
8 # machine learning model loading
9 MODEL_PATH = 'expense_predictor_model.pkl'
10 model = None
11
12 if os.path.exists(MODEL_PATH):
13     model = joblib.load(MODEL_PATH)
14     print(f'Machine learning model loaded successfully from {MODEL_PATH}')
15 else:
16     print(f'WARNING: Model file {MODEL_PATH} not found. Please run train_model.py first')
17
18 @app.route('/')
19 def home():
20     return jsonify({'Message': "Welcome to Fatima's Smart Business Expense Predictor API"})
21
22
23
24
25 #Run the flask Application
26 if __name__ == '__main__':
27     app.run(debug = True)
```

```
Command Prompt - python  +
C:\Users\Lenovo-User\Desktop>cd expense_predictor_api
C:\Users\Lenovo-User\Desktop\expense_predictor_api>venv\scripts\activate
(venv) C:\Users\Lenovo-User\Desktop\expense_predictor_api>python train_model.py
--- Training Smart Business Expense Predictor Model ---
Data loaded successfully. Shape: (36, 1)
Train data size: 36 samples.
Model Training Complete
Model Coefficient (slope): 182.38 R/month
Model Intercept: 10236.99
Model Evaluation
Mean Absolute Error: 862.83
R-Squared score: 0.79
Machine learning model saved!
(venv) C:\Users\Lenovo-User\Desktop\expense_predictor_api>python app.py
WARNING: Model file expense_predictor_model.pkl not found. Please run train_model.py first
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
WARNING: Model file expense_predictor_model.pkl not found. Please run train_model.py first
* Debugger is active!
* Debugger PIN: 193-513-701
127.0.0.1 - - [24/Mar/2026 09:56:02] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [24/Mar/2026 09:56:02] "GET /favicon.ico HTTP/1.1" 404 -
```



```

22 @app.route('/predict_expense', methods = ['POST'])
23 def predict_expense():
24     #check if the model was loaded successfully
25     if model is None:
26         return jsonify({'error': 'Expense prediction model not available. Please train the model first'}), 500
27     request_data = request.get_json()
28
29     if not request_data or 'month_number' not in request_data:
30         return jsonify({'error': 'Please provide "month_number" in the request body'}), 400
31     try:
32         month_number = int(request_data['month_number'])
33
34         if month_number <=0:
35             return jsonify({'error': '"month_number" must be positive integer'}), 400
36         prediction_input = np.array([[month_number]])
37
38         predicted_expense = model.predict(prediction_input) [0]
39         predicted_expense = max(0, round(predicted_expense, 2))
40
41         return jsonify({
42             'month_number': month_number,
43             'predicted_expense_R': predicted_expense
44         })
45     except ValueError:
46         return jsonify ({'error': 'invalid 'month_number' value. Must be an integer' }), 400
47     except Exception as e:
48         return jsonify({'error': f' An unexpected error ocured during prediction: {str(e)}'}, 500
49
50 #Run the flask Application
51 if __name__ == '__main__':
52     app.run(debug =True)
53

```

```

File "C:\Users\Lenovo-User\Desktop\expense_predictor_api\app.py", line 34
    if month_number <= 0:
    ^
IndentationError: unindent does not match any outer indentation level

(venv) C:\Users\Lenovo-User\Desktop\expense_predictor_api>python app.py
File "C:\Users\Lenovo-User\Desktop\expense_predictor_api\app.py", line 34
    if month_number <= 0:
    ^
IndentationError: unindent does not match any outer indentation level

(venv) C:\Users\Lenovo-User\Desktop\expense_predictor_api>python app.py
WARNING: Model file expense_predictor_model.pkl not found. Please run train_model.py first
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production web server.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
WARNING: Model file expense_predictor_model.pkl not found. Please run train_model.py first
* Debugger is active!
* Debugger PIN: 193-513-701

```

Smart Business Expense Predictor

Enter Month Number

Predict Expense

Prediction for Month 37:
Estimated Expense: R 17237.55